



SCEGGS, DARLINGHURST

Mathematics

3 Unit (Additional)
and
3/4 Unit (Common)

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16

2000

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

TIME ALLOWED : 2 Hours
(+ 5 minutes reading time)

INSTRUCTIONS:

- Attempt ALL SEVEN questions and show all necessary working.
- Marks will be deducted for careless or badly arranged work.
- ALL questions are of equal value.
- START EACH QUESTION ON A NEW PAGE.
- Make sure your student number is on each page.
- Approved calculators and templates may be used.
- Standard Integrals are printed on the last page. These may be removed for your convenience.

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or format of the Higher School Certificate Examination.

Question 1 : (12 Marks)

- (a) Solve : $\frac{2x}{x+1} \leq 1$ 3
- (b) Consider each different arrangement of the letters of the word INFINITE. 3
- How many different words are possible?
 - If one of these words is chosen at random, what is the probability that the 3 I's are together?
- (c) Explain how you could find the coordinates of the point C that divides the interval joining A(1, 4) to B(-2, 10) in the ratio 1 : 5 without using a formula. 2
- (d) Give an example of a value of x in radians for which $\sin^{-1}(\sin x) \neq x$ 1
- (e) Prove that : 3
- $$n! + (n-1)! + (n-2)! = n^2(n-2)!$$

Question 2 : Start a new page (12 Marks)

- (a) Find: $\int \cos^2 5x \, dx$ 2
- (b) Find the exact volume of the solid of revolution formed when the area between the curve $y = \frac{1}{\sqrt{x^2 + 9}}$, the x axis and the lines $x = 0$ and $x = 3\sqrt{3}$ is rotated about the x axis. 3
- (c) Evaluate $\int_0^1 \frac{4x}{(4x+1)^2} \, dx$ using the substitution $u = 4x+1$ 4
- (d) Find $\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 2x}$ 3

Question 3 : Start a new page (12 Marks)

(a) Let $f(x) = x^3 + 3x^2 - 10x - 24$

3

(i) Calculate $f(-2)$

(ii) Hence, express $f(x)$ as the product of three linear factors.

(b) Let α, β and γ be the roots of the equation $x^3 - 3x + 5 = 0$

4

Find the values of :

(i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta\gamma$

(iii) $(\alpha - 1)(\beta - 1)(\gamma - 1)$

(c) Consider the parabola $x^2 = 4ay$

5

(i) Show that the equation of the normal to this parabola at the point $P(2ap, ap^2)$ is given by $x + py = ap^3 + 2ap$.(ii) If this normal meets the parabola again at $Q(2aq, aq^2)$, show that $p^2 + pq + 2 = 0$.

Questions continue over ...

Question 4 : Start a new page (12 Marks)

(a) Find the coefficient of x^3 in $\left(3x^3 + \frac{1}{x}\right)^9$

5

(b) A function is defined as $f(x) = 1 + e^{2x}$

(i) Write down the range of this function.

(ii) Show that the inverse function can be defined as $f^{-1}(x) = \frac{1}{2} \ln(x - 1)$ (iii) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ (iv) The equation of the normal to the curve $y = f^{-1}(x)$ at the point where $f^{-1}(x) = 0$ is given by the equation $2x + y - 4 = 0$. Show that the point of intersection of this normal and $y = f(x)$ can be derived from the equation $e^{2x} + 2x = 3$.(v) By taking $x = 0.4$ as the first approximation of the root to $e^{2x} + 2x = 3$, use one application of Newton's Method to find a better approximation of the root, correct to 3 significant figures.

Questions continue over ...

Question 5 : Start a new page (12 Marks)

- (a) Solve $2^{2x+1} - 5(2^x) + 2 = 0$ 3
- (b) Prove that $2^{10n+3} + 3$ is divisible by 11 for all non-negative integers n by Mathematical Induction. 4
- (c) A particle moves in a straight line with Simple Harmonic Motion. At time t seconds, its displacement x metres from a fixed point O is given by: 5
- $$x = 5 \sin \frac{\pi}{2} \left(t + \frac{1}{3} \right)$$
- (i) Show that $\ddot{x} = -\frac{\pi^2}{4}x$
- (ii) State the period and the amplitude of the motion.
- (iii) Find the magnitude of the acceleration when $x = 2\frac{1}{2}$

Questions continue over ...

Question 6 : Start a new page (12 Marks)

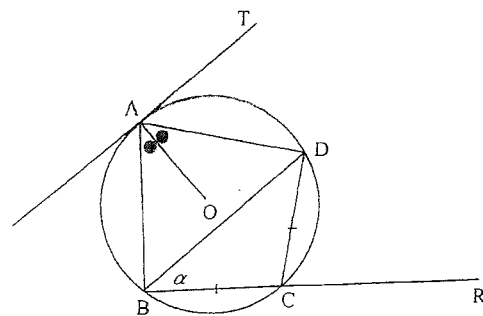
- (a) N is the number of aardvarks in a certain population at time t years. The population size N satisfies the equation $\frac{dN}{dt} = -k(N - 1000)$, for some constant k . 6
- (i) Verify by differentiation that $N = 1000 + Ae^{-kt}$ (where A is a constant) is a solution of the equation, $\frac{dN}{dt} = -k(N - 1000)$.
- (ii) Initially there are 2500 aardvarks but after 2 years there are only 2200 left. Find the values of A and k .
- (iii) Sketch the graph of population size against time.
- (b) During the Euro2000 soccer tournament, Brett is standing 25 metres away from the goal line. He kicks a soccer ball off the ground at an angle of 30° to the horizontal with an initial velocity of V m/s. The ball hits the top bar which is 2.4 metres directly above the goal line. Neglecting air resistance and assuming that acceleration due to gravity is 10m/s^2 , find: 6
- (i) the horizontal and vertical components of the displacement of the ball in terms of the initial velocity, V .
- (ii) the Cartesian equation of the motion for the path of the ball.
- (iii) the initial velocity of the ball, correct to 1 decimal place.

Questions continue over ...

Question 7 : Start a new page (12 Marks)

- (a) (i) Solve $(2x-1)(2x-\sqrt{3}) < 0$ 3
- (ii) Hence solve $(2\sin\theta-1)(2\sin\theta-\sqrt{3}) < 0$ for $0 \leq \theta \leq 2\pi$

- (b) Points A, B, C and D lie on a circle centre O . The line TA is a tangent to the circle at A , and BC is produced to R . The interval OA bisects $\angle BAD$, and $BC = CD$. 4
- The size of $\angle DBC$ is α .



NOT TO SCALE

Copy or trace the diagram.

- (i) Explain why $\angle DCR = 2\alpha$
- (ii) Show that $\angle OAD = \alpha$
- (iii) Prove that $\angle ABC$ is a right angle.
- (c) (i) Write down the formula for the coefficient of x^r in the expansion of $(1+x)^n$, where r and n are positive integers and $1 \leq r \leq n$ 5
- (ii) Let s and t be positive consecutive integers with $t = s + 1$. Show that $s^{2n} + 2nt - 1$ is divisible by t^2 .
- (iii) Hence find a perfect square that is a factor of $5^{20} + 119$.

- END OF EXAMINATION -

3 UNIT MATHEMATICS

2000 TRIAL SOLUTIONS

QUESTION 1: 12 marks

$$\frac{2x}{x+1} \leq 1 \quad (x \neq -1)$$

$$2x(x+1) \leq (x+1)^2$$

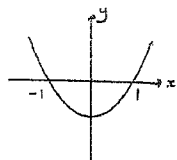
$$2x^2 + 2x \leq x^2 + 2x + 1$$

$$\therefore x^2 - 1 \leq 0$$

$$\therefore -1 \leq x \leq 1$$

$$x \neq -1$$

$$\therefore -1 < x \leq 1$$

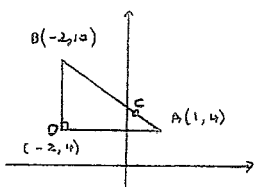


i) INFINITE

$$\text{Number of different words} = \frac{8!}{3!2!} = 3360$$

$$\text{ii) Number of possibilities with I's together} = \frac{6!}{2!} = 360$$

$$\therefore \text{Probability} = \frac{360}{3360} = \frac{3}{28}$$



is right angled triangle.

is $\frac{1}{5}$ of length AB and subtract this distance from the x-value of A - x value.

is $\frac{1}{5}$ of length BD and add this distance to the y-value of A - y value.

$$\text{(e) LHS} = n! + (n-1)! + (n-2)!$$

$$= (n-2)! [n(n-1) + (n-1) + 1]$$

$$= (n-2)! (n^2 - n + n - 1 + 1)$$

$$= n^2 (n-2)!$$

$$= \text{RHS}$$

QUESTION 2: 12 marks

$$\text{(a) } \int \cos^2 5x \, dx$$

$$= \frac{1}{2} \int \cos 10x + 1 \, dx$$

$$= \frac{1}{2} \left[\frac{\sin 10x}{10} + x \right] + C$$

$$= \frac{1}{20} \sin 10x + \frac{1}{2} x + C$$

$$\text{(b) } V = \pi \int_0^{3\sqrt{3}} \frac{1}{x^2 + 9} \, dx$$

$$= \left[\frac{\pi}{3} + \arctan \frac{x}{3} \right]_0^{3\sqrt{3}}$$

$$= \frac{\pi}{3} \tan^{-1} \sqrt{3} - \frac{\pi}{3} \tan^{-1} 0$$

$$= \frac{\pi^2}{9}$$

$$\text{(c) } \int_0^1 \frac{4x}{(4x+1)^2} \, dx \quad u = 4x+1$$

$$\frac{du}{dx} = 4$$

$$\text{when } x=0, u=1$$

$$x=1, u=5$$

$$= \int_1^5 \frac{u-1}{u^2} \cdot \frac{1}{4} \, du$$

$$= \frac{1}{4} \int_1^5 \frac{1}{u} - \frac{1}{u^2} \, du$$

$$= \frac{1}{4} \left[\ln u + \frac{1}{u} \right]_1^5$$

$$= \frac{1}{4} \left[\left(\ln 5 + \frac{1}{5} \right) - \left(\ln 1 + 1 \right) \right]$$

$$= \frac{1}{4} \left(\ln 5 - \frac{4}{5} \right)$$

$$= \frac{1}{4} \ln 5 - \frac{1}{5}$$

$$\text{(d) } \lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 2x} = \lim_{x \rightarrow 0} \frac{2 \sin 2x \cos 2x}{\frac{\sin 2x}{\cos 2x}} = \lim_{x \rightarrow 0} 2 \cos^2 2x = 2$$

QUESTION 3: 12 marks

$$\text{(a) } f(x) = x^3 + 3x^2 - 10x - 24$$

$$\text{(i) } f(-2) = -8 + 12 + 20 - 24 = 0$$

$$\text{(ii) } \therefore f(x) = (x+2)(x^2 + x - 12)$$

$$= (x+2)(x+4)(x-3)$$

$$\text{(b) } x^3 - 3x + 5 = 0$$

$$\text{(i) } \alpha + \beta + \gamma = 0$$

$$\text{(ii) } \alpha\beta\gamma = -5$$

$$\text{(iii) } (\alpha-1)(\beta-1)(\gamma-1)$$

$$= (\alpha\beta - \alpha - \beta + 1)(\gamma - 1)$$

$$= \alpha\beta\gamma - \alpha\gamma - \beta\gamma + \gamma - \alpha\beta + \alpha + \beta - 1$$

$$= \alpha\beta\gamma - (\alpha\beta + \beta\gamma + \gamma\alpha) + (\alpha + \beta + \gamma) - 1$$

$$= -5 + 3 + 0 - 1$$

$$= -3$$

$$\text{(c) (i) } x^2 = 4ay$$

$$\therefore y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$\text{At } P, \frac{dy}{dx} = \frac{2ap}{2a} = p$$

$$\therefore \text{Grad norm} = -\frac{1}{p}$$

$\therefore$ Eqn normal:

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\therefore x + py = ap^3 + 2ap$$

(ii) Q lies on normal:

$$2aq + paq^2 = ap^3 + 2ap$$

$$\therefore 2q + pq^2 = p^3 + 2p$$

$$p^3 - pq^2 + 2p - 2q = 0$$

$$p(p^2 - q^2) + 2(p - q) = 0$$

$$(p - q)(p(p + q) + 2) = 0$$

but $p \neq q$ since P and Q are distinct points

$$\therefore (p - q) \neq 0$$

$$\therefore p^2 + pq + 2 = 0$$

QUESTION 4: 12 marks

$$\text{(a) } \left(3x^2 + \frac{1}{x} \right)^9$$

$$T_{k+1} = {}^9C_k (3x^2)^{9-k} \cdot \left(\frac{1}{x} \right)^k$$

$$= {}^9C_k \cdot 3^{9-k} \cdot x^{18-2k} \cdot x^{-k}$$

$$= {}^9C_k \cdot 3^{9-k} \cdot x^{18-3k}$$

$\therefore$ Coefficient of x^3 :

$$\Rightarrow 18 - 3k = 3$$

$$3k = 15$$

$$k = 5$$

$$\therefore \text{Coefficient} = {}^9C_5 \times 3^{9-5}$$

$$= 126 \times 3^4$$

$$= 10206$$

$$\text{(b) } f(x) = 1 + e^{2x}$$

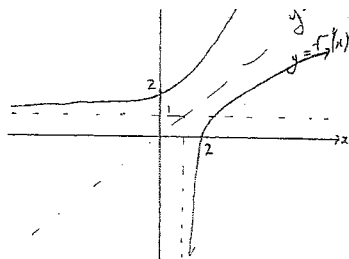
$$\text{(i) } y > 1$$

$$\text{(ii) } x = 1 + e^{2y}$$

$$\therefore x - 1 = e^{2y}$$

$$\ln(x-1) = 2y$$

$$\therefore y = \frac{1}{2} \ln(x-1)$$



(iv) Normal at (2, 0) is

$$2x + y - 4 = 0$$

$$y = 4 - 2x$$

$$4 - 2x = 1 + e^{2x}$$

$$e^{2x} + 2x = 3$$

(v) $f(x) = e^{2x} + 2x - 3$

$$f'(x) = 2e^{2x} + 2$$

$$f(0.4) = 0.025 \dots$$

$$f'(0.4) = 6.451 \dots$$

$$\therefore a_1 = 0.4 - \frac{0.025}{6.451}$$

$$= 0.396 \text{ (correct to 3 significant figures)}$$

$$(a) 2^{2x+1} - 5(2^x) + 2 = 0$$

$$2(2^{2x}) - 5(2^x) + 2 = 0$$

$$\text{let } u = 2^x$$

$$2u^2 - 5u + 2 = 0$$

$$(2u - 1)(u - 2) = 0$$

$$\therefore u = \frac{1}{2} \text{ or } 2$$

$$\therefore x = -1 \text{ or } 1$$

(b) $n = 0$:

$$\text{LHS: } 2^3 + 3 = 11 \text{ which is divisible by 11.}$$

Assume true for $n = k$:

$$\text{i.e. } 2^{10k+3} + 3 = 11M \text{ for some } M \in \mathbb{Z}^+$$

When $n = k+1$:

$$2^{10(k+1)+3} + 3$$

$$= 2^{10k+10+3} + 3$$

$$= 2^{10} \cdot 2^{10k+3} + 3$$

$$= 2^{10} (11M - 3) + 3 \text{ by induction hypothesis}$$

$$= 11M \cdot 2^{10} - 3 \cdot 2^{10} + 3$$

$$= 11M \cdot 2^{10} - 3069$$

$$= 11 (2^{10} M - 279)$$

which is divisible by 11

since $2^{10} M > 279$

for all $M \in \mathbb{Z}^+$.

$\therefore$ If hypothesis is true for $n = k$, it is also true for $n = k+1$.

Since it is true for $n = 0$, it is also true for $n = 1, 2, \dots$ and hence all non-negative integers by mathematical induction.

$$(i) \dot{x} = \frac{5\pi}{2} \cos \frac{\pi}{2} (t + \frac{1}{3})$$

$$\ddot{x} = -\frac{5\pi^2}{4} \sin \frac{\pi}{2} (t + \frac{1}{3})$$

$$= -\frac{\pi^2}{4} x$$

(ii) Amp = 5

$$\text{Period} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

$$(iii) \ddot{x} = -\frac{\pi^2}{4} x \cdot \frac{5}{2} = -\frac{5\pi^2}{8} x$$

QUESTION 6: 12 marks

(a) (i) $N = 1000 + Ae^{-kt}$

$$\frac{dN}{dt} = -k \cdot Ae^{-kt} = -k(N - 1000)$$

(ii) when $t = 0$, $N = 2500$

$$2500 = 1000 + Ae^0$$

$$\therefore A = 1500$$

when $t = 2$, $N = 2200$

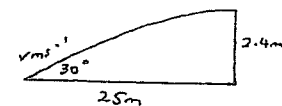
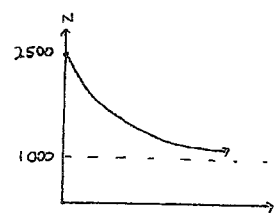
$$2200 = 1000 + 1500e^{-2k}$$

$$0.8 = e^{-2k}$$

$$\therefore k = -\frac{1}{2} \ln 0.8$$

$$= 0.11157$$

(iii) $N = 1000 + 1500e^{-0.11t}$



$$v \sin 30 = \frac{x}{2} \Rightarrow v \cos 30 = \frac{\sqrt{3}v}{2}$$

(i) Horizontal:

$$\ddot{x} = 0$$

$$\therefore \dot{x} = c$$

$$\text{when } t = 0, \dot{x} = \frac{\sqrt{3}v}{2}$$

$$\therefore \dot{x} = \frac{\sqrt{3}v}{2}$$

$$x = \frac{\sqrt{3}}{2} vt + c$$

$$\text{when } t = 0, x = 0$$

$$\therefore x = \frac{\sqrt{3}}{2} vt$$

Vertical:

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$\text{when } t = 0, \dot{y} = \frac{v}{2}$$

$$\therefore \dot{y} = -10t + \frac{v}{2}$$

$$\therefore y = -5t^2 + \frac{vt}{2} + c$$

$$\text{when } t = 0, y = 0$$

$$\therefore y = -5t^2 + \frac{vt}{2}$$

(ii) $t = \frac{2x}{\sqrt{3}v}$

$$\therefore y = -5 \left(\frac{2x}{\sqrt{3}v} \right)^2 + \frac{v}{2} \left(\frac{2x}{\sqrt{3}v} \right)$$

$$y = -\frac{20x^2}{3v^2} + \frac{x\sqrt{3}}{3}$$

(iii) when $x = 25$, $y = 2.4$

$$2.4 = -\frac{20 \times 25^2}{3v^2} + \frac{25\sqrt{3}}{3}$$

$$7.2v^2 = -12500 + 25\sqrt{3}v^2$$

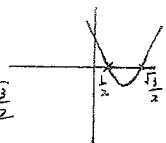
$$\therefore v^2 = \frac{12500}{25\sqrt{3} - 7.2}$$

$$= 346.248 \dots$$

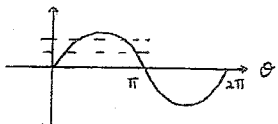
$$\therefore v = 18.6 \text{ m/s}$$

$$(i) (2x-1)(2x-\sqrt{3}) < 0$$

$$\therefore \frac{1}{2} < x < \frac{\sqrt{3}}{2}$$



$$ii) \therefore \frac{1}{2} < \sin \theta < \frac{\sqrt{3}}{2}$$



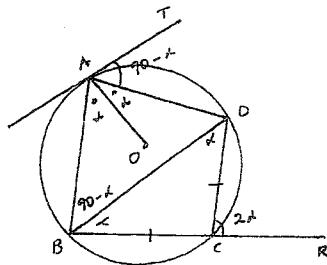
$$\sin \theta = \frac{1}{2}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

$$\therefore \frac{\pi}{6} < \theta < \frac{\pi}{3} \text{ or } \frac{2\pi}{3} < \theta < \frac{5\pi}{6}$$



$\angle BDC = \alpha$ ($\angle$ opp sides in Δ are α)

$\therefore \angle ODC = 2\alpha$ (exterior $\angle$ in Δ = 2 interior opp $\angle$)

$\therefore \angle BAD = 2\alpha$ (exterior $\angle$ in cyclic quad = opp $\angle$)

$\therefore \angle OAD = \alpha$ (OA bisects $\angle BAD$ - given).

$\therefore \angle OAT = 90^\circ$ ($\angle$ between tangent + radius = 90°)

$\therefore \angle DAT = 90^\circ - \alpha$

$\therefore \angle ABD = 90^\circ - \alpha$ ($\angle$ between chord + tangent = $\angle$ in the alt. segment)

$\therefore \angle ABC = 90^\circ - \alpha + \alpha = 90^\circ$

$\therefore \angle ABC$ is a right angle.

$$(c) (i) \text{ coefficient} = {}^nC_r = \frac{n!}{(n-r)!r!}$$

$$(ii) 5^{2n} + 2nt - 1$$

$$= (t-1)^{2n} + 2nt - 1$$

$$= (t^{2n} + {}^{2n}C_1 t^{2n-1} (-1)^1 +$$

$${}^{2n}C_2 t^{2n-2} (-1)^2 + \dots +$$

$${}^{2n}C_{2n-1} t (-1)^{2n-1} + (-1)^{2n}) + 2nt - 1$$

$$= (t^{2n} - 2nt^{2n-1} + {}^{2n}C_2 t^{2n-2} - \dots$$

$$+ {}^{2n}C_{2n-2} t^2 - 2nt + 1) + 2nt - 1$$

$$= t^{2n} - 2nt^{2n-1} + {}^{2n}C_2 t^{2n-2} + \dots + {}^{2n}C_{2n-2} t^2$$

$$= t^2 (t^{2n-2} - 2nt^{2n-3} + \dots + {}^{2n}C_2)$$

$\therefore 5^{2n} + 2nt - 1$ is divisible by t^2

$$(iii) 5^{20} + 119$$

$$= 5 \therefore t = 6$$

$$n = 10$$

$\therefore$ By (ii) $5^{20} + 120 - 1$

is divisible by $t^2 = 36$.

END OF SOLUTIONS.